Query Syntax & Semantics

Syntax summary

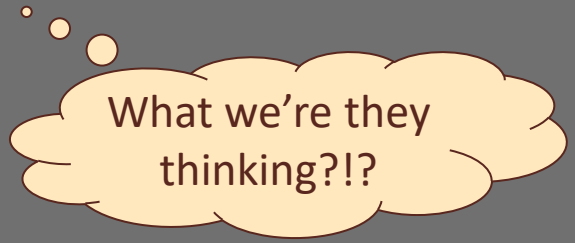
```
SELECT [DISTINCT] Computations
FROM      Table0
   INNER JOIN Table1 ON JoinCond1
   …
WHERE     FilterCond
ORDER BY  Computations
LIMIT     n;
```

$\{\text{Computations} \mid \text{Table}(t) \land \text{JoinCond} \land \text{FilterCond}\}$

Semantics summary

```
SELECT [DISTINCT] Computations
FROM      Table0
   INNER JOIN Table1 ON JoinCond1
   …
WHERE     FilterCond
ORDER BY  Computations
LIMIT     n;
```

3. Computations, [eliminate duplicates]
1. Get & join data
2. Filter rows
4. Sort
5. Use first n rows



$\{Computations \mid Table(t) \wedge JoinCond \wedge FilterCond\}$

Example

```
SELECT   p.purch_product,
         p.purch_price * p.purch_quant AS total
FROM     Purchase AS p
WHERE    p.purch_price * p.purch_quant > 1::MONEY
ORDER BY total
LIMIT    1
```

3. Computations

1. Get & join data
2. Filter rows
4. Sort
5. Use first n rows